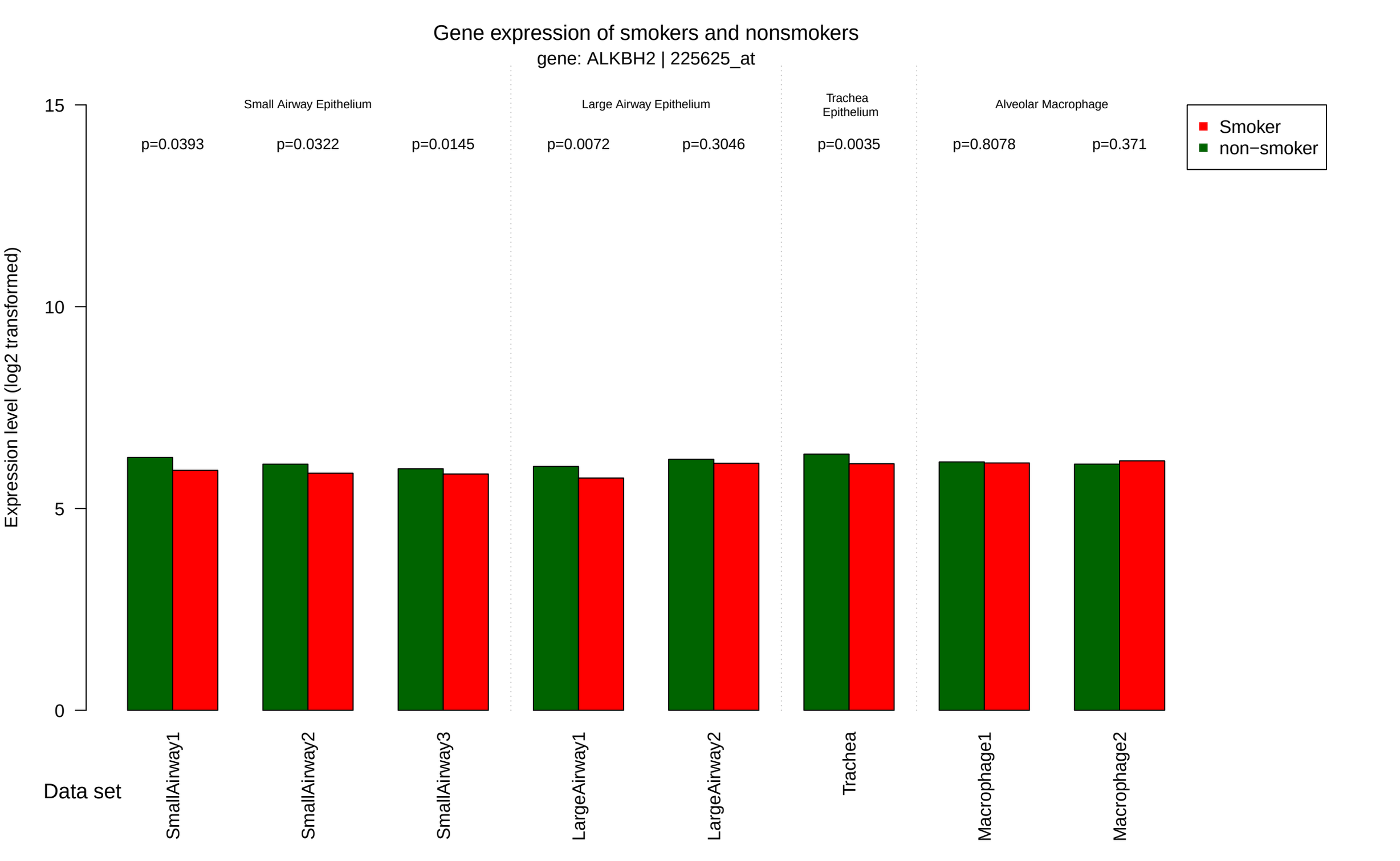
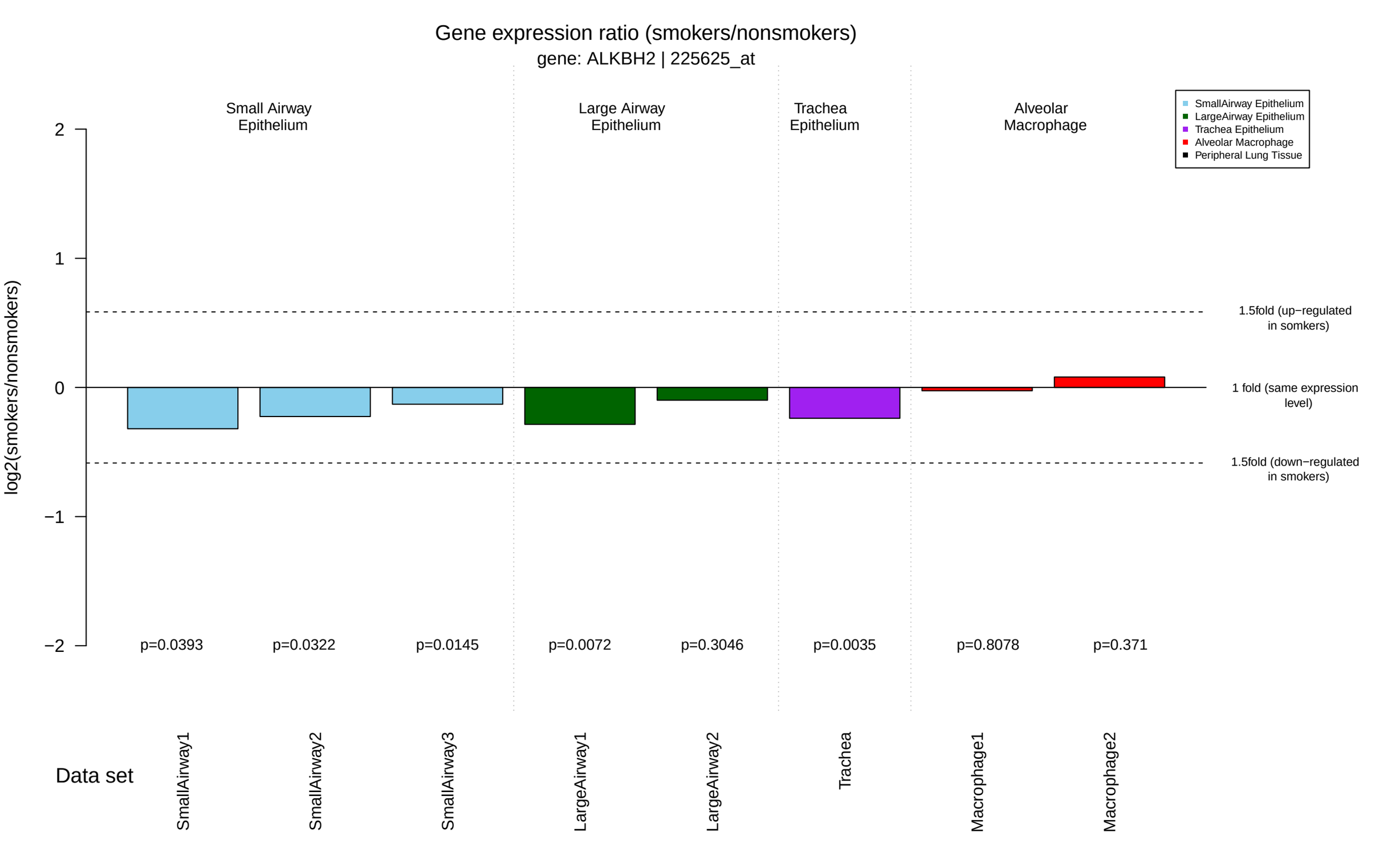
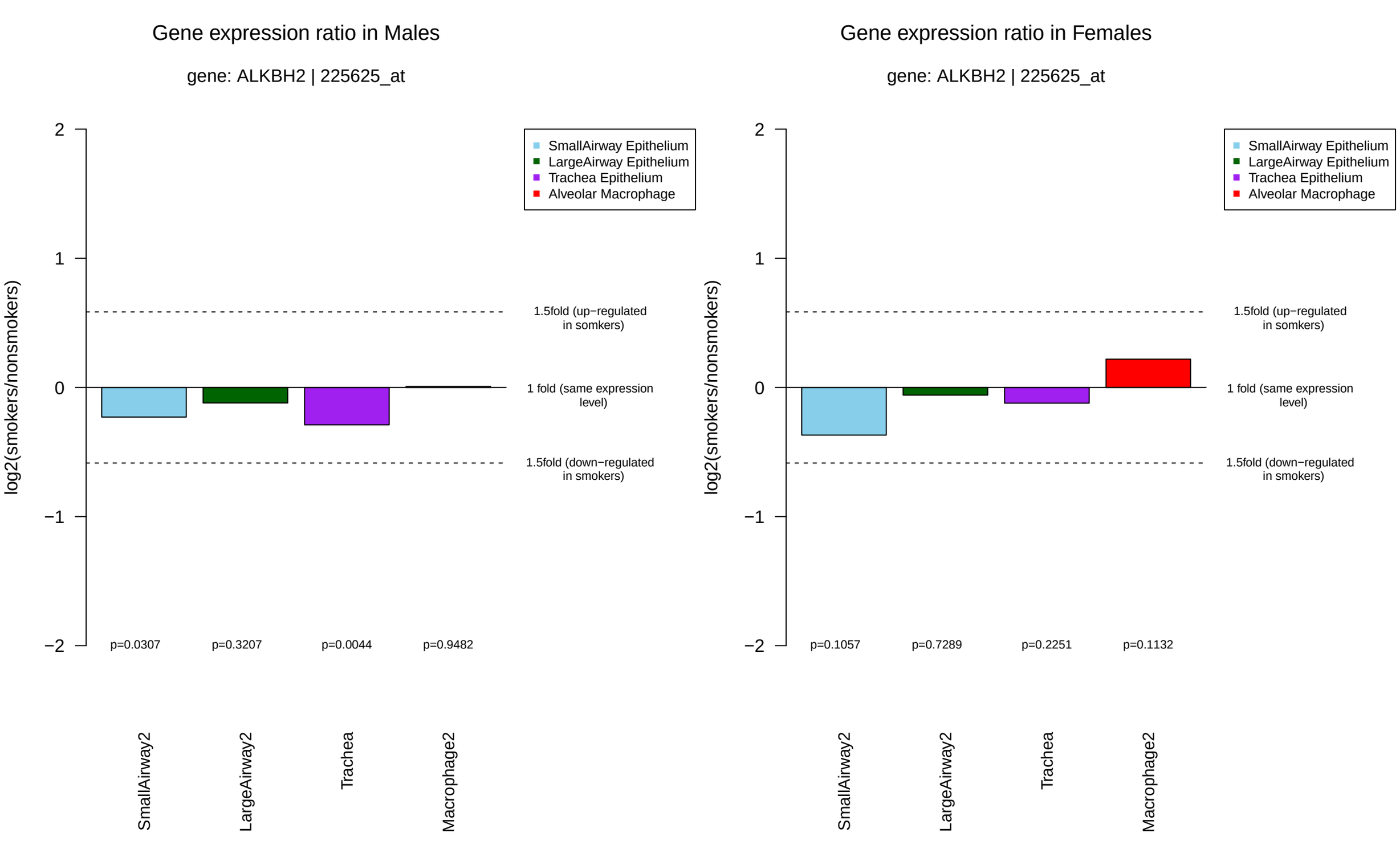
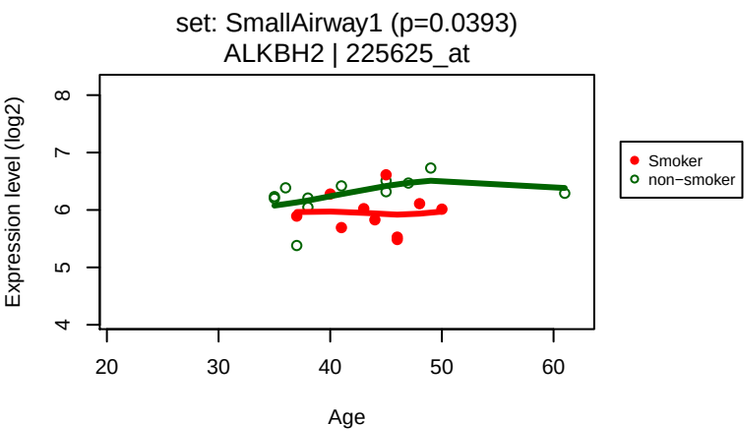
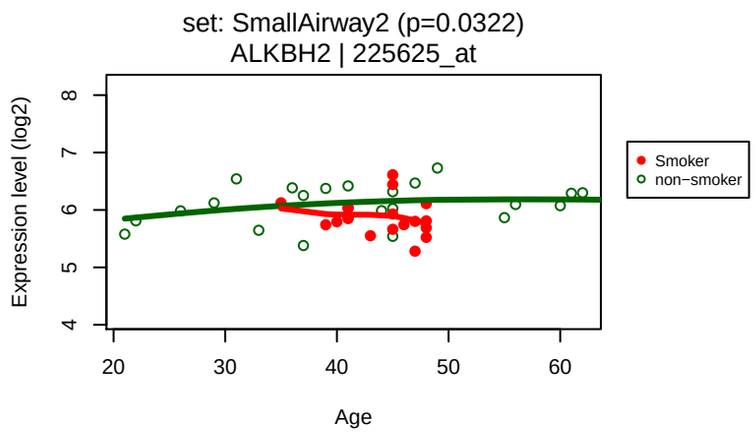
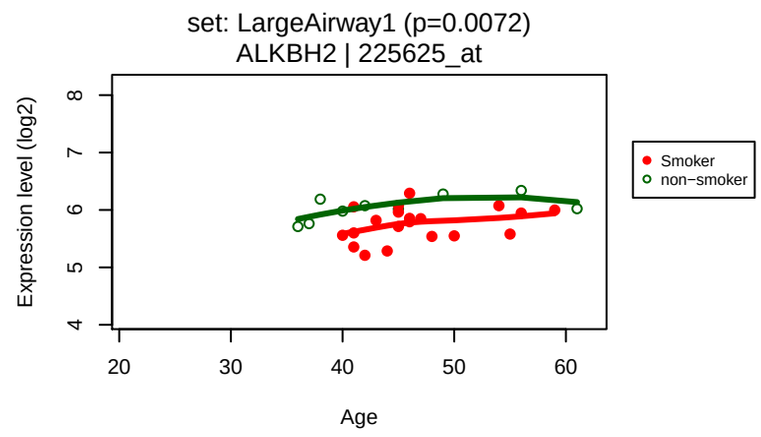

Gene expression of smokers and nonsmokers
gene: ALKBH2 | 225625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0393
p=0.0322
p=0.0145
p=0.0072
p=0.3046
p=0.0035
p=0.8078
p=0.371
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

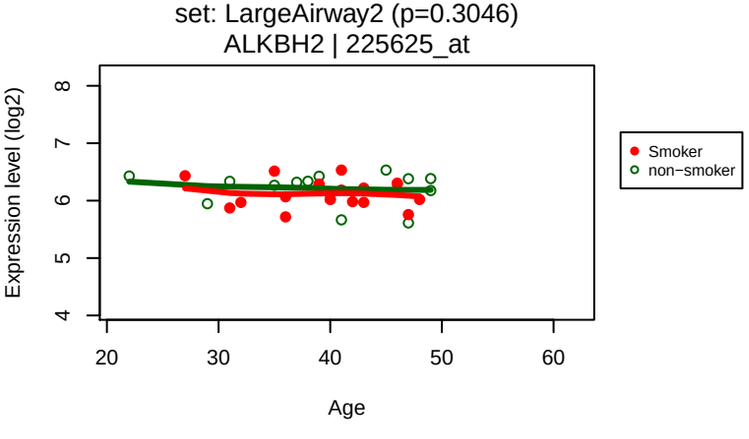

Gene expression ratio (smokers/nonsmokers)
gene: ALKBH2 | 225625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0393
p=0.0322
p=0.0145
p=0.0072
p=0.3046
p=0.0035
p=0.8078
p=0.371
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

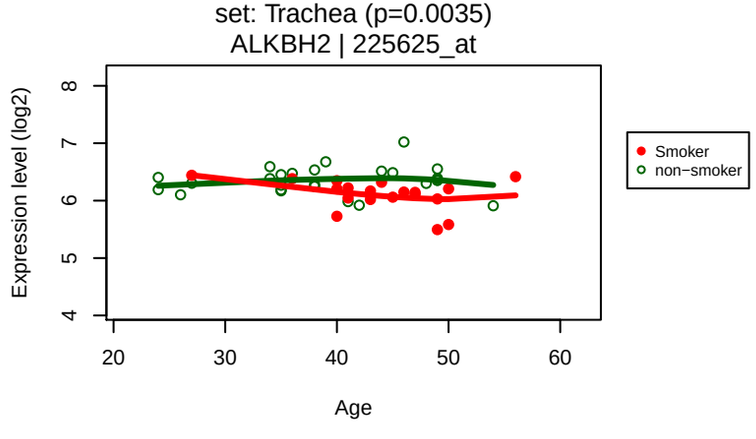

Gene expression ratio in Males
gene: ALKBH2 | 225625_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0307
p=0.3207
p=0.0044
p=0.9482
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALKBH2 | 225625_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1057
p=0.7289
p=0.2251
p=0.1132
SmallAirway2
LargeAirway2
Trachea
Macrophage2

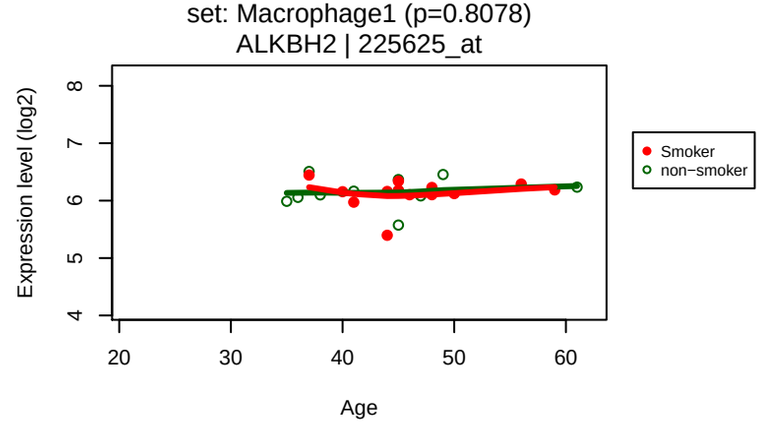

set: SmallAirway1 (p=0.0393)
ALKBH2 | 225625_at
Expression level (log2)
Age
Smoker
non−smoker

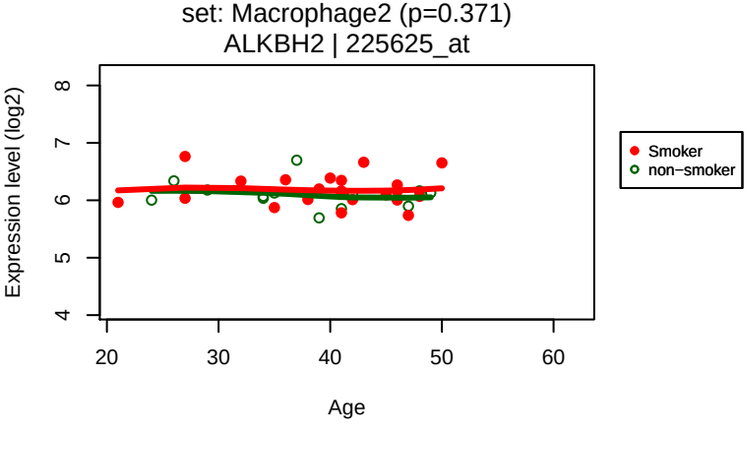

set: SmallAirway2 (p=0.0322)
ALKBH2 | 225625_at
Expression level (log2)
Age
Smoker
non−smoker

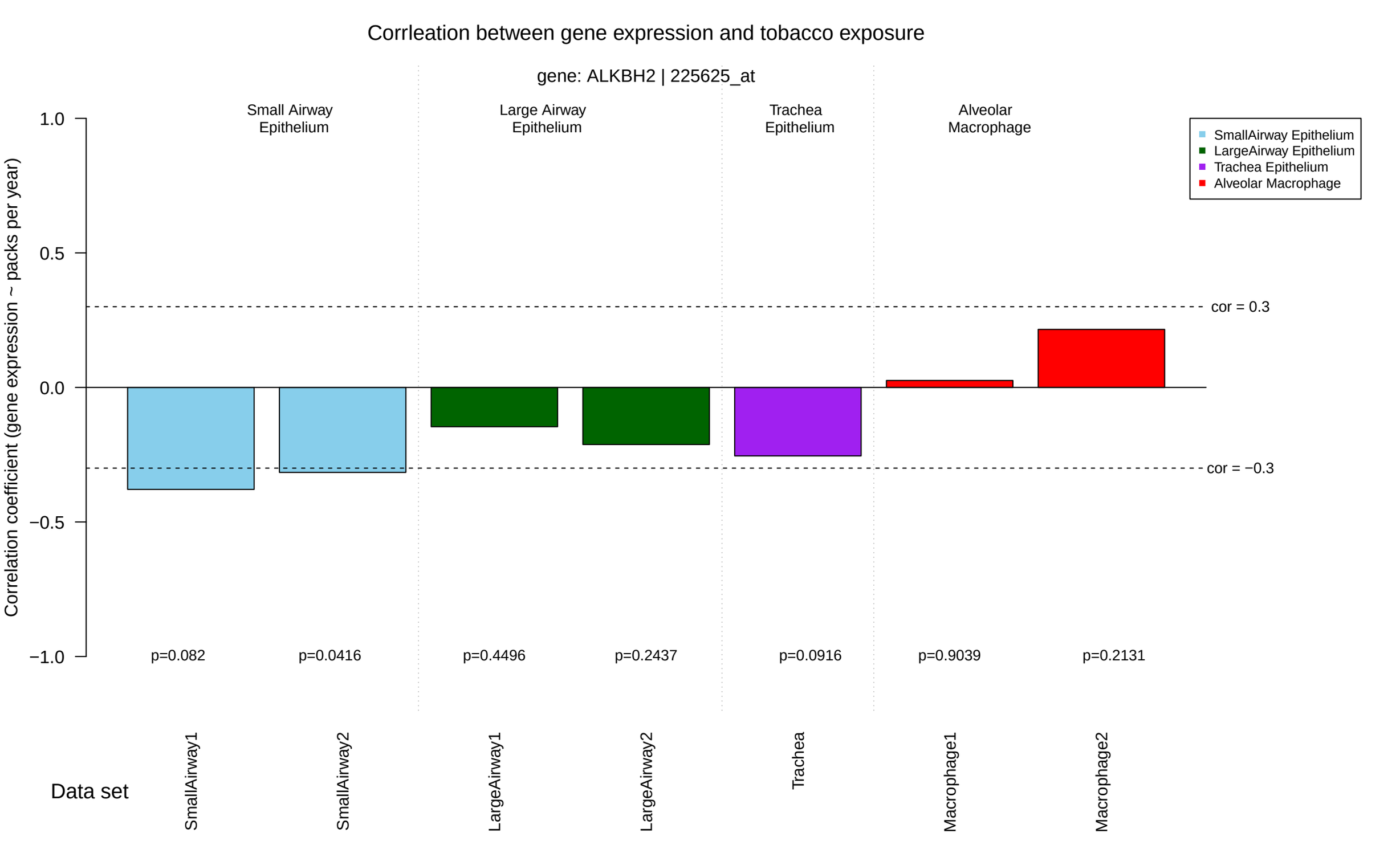

set: LargeAirway1 (p=0.0072)
ALKBH2 | 225625_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3046)
ALKBH2 | 225625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0035)
ALKBH2 | 225625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8078)
ALKBH2 | 225625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.371)
ALKBH2 | 225625_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALKBH2 | 225625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.082
p=0.0416
p=0.4496
p=0.2437
p=0.0916
p=0.9039
p=0.2131
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2